'formatting' to the sources also. This is not really necessary from the point of view of LaTeX since the output of the last example is he same even if we input

```
\begin{tabbing}
Row 1 Column 1\hspace{2cm}\=Row 1 Column 2\\[5pt]
\>Row 2 Column 2\hspace{1.5cm}\=Row 2 Column 3\+\+\\[5pt]
Row 3 Column3\-\\[5pt]
Row 4 Column 2\>Row 4 Column 3\\[5pt]
\<Row 5 Column\>Row 5 Column 2\>Row 5 Column 3\\[5pt]
Row 6 Column 2\>Row 6 Column 3\-\\[5pt]
Row 7 Column 1\>Row 7 Column 2\>Row 7 Column 3\\[5pt]
Row 8 Column 1\`\textbf{Flush right}\\[5pt]
Row 9 Column 1\>and\'Row 9 Column 2\\[5pt]
\pushtabs
 Row 10 New Column 1\hspace{2.5cm}\=Row 10 New Column 2\\[5pt]
 Row 11 New Column 2\>Row 11 New Column 2\\[5pt]
\poptabs
Row 12 Old Column 1\>Row 12 Old Column 2\>Row 12 Old Column 3
\end{tabbing}
```

LaTeX can make sense out of this, but we humans cannot. And such a jumble makes editing a hopeless task. The moral? Keep the source (humanly) readable.

VII.2. TABLES

Another way to format text into columns and rows is to use the `tabular` environment. Let's see it in action by means of an example.

The table below shows the sizes of the planets of our solar system.

Planet	Diameter(km)
Mercury	4878
Venus	12104
Earth	12756
Mars	6794
Jupiter	142984
Saturn	120536
Uranus	51118
Neptune	49532
Pluto	2274

As can be seen, Pluto is the smallest and Jupiter the largest

Now look at the source of this output

```
The table below shows the sizes of the planets of our solar system.
\begin{center}
  \begin{tabular}{lr}
    Planet  & Diameter(km)\\[5pt]
    Mercury & 4878\\
    Venus   & 12104\\
    Earth   & 12756\\
    Mars    & 6794\\
    Jupiter & 142984\\
    Saturn  & 120536\\
```

```
    Uranus  & 51118\\
    Neptune & 49532\\
    Pluto   & 2274
  \end{tabular}
\end{center}
As can be seen, Pluto is the smallest and Jupiter the largest
```

The `\begin{center} ... \end{center}` commands centralize the table. The table itself is produced by the `\begin{tabular} ...\end{tabular}` commands. The `{lr}` specification immediately after the `\begin{tabular}` indicates there are two *columns* in the table with the entries in the first column aligned on the *left* and the entries in the second column aligned on the *right*. The entries in each column are separated by the `&` symbol and the terminatio of each row is signalled by the `\\` symbol. The `\\[5pt]` after the first row specifies as usual, an additional vertical space of 5 points after this row in the output.

In addition to the column specifiers `l` and `r` we also have a specifier `c` which makes the entries in the corresponding column *centrally* aligned. For example the input

```
\begin{center}
  \begin{tabular}{cr}
    Planet  & Diameter(km)\\[5pt]
    Mercury & 4878\\
    Venus   & 12104\\
    Earth   & 12756\\
    Mars    & 6794\\
    Jupiter & 142984\\
    Saturn  & 120536\\
    Uranus  & 51118\\
    Neptune & 49532\\
    Pluto   & 2274
  \end{tabular}
\end{center}
```

produces the output below

Planet	Diameter(km)
Mercury	4878
Venus	12104
Earth	12756
Mars	6794
Jupiter	142984
Saturn	120536
Uranus	51118
Neptune	49532
Pluto	2274

There's yet another column specifier `p` which allows us to set column entries in a *box* of specified width (technically a "parbox"—see Chapter X). Suppose you want something like this

Planet	Features
Mercury	Lunar like crust, crustal faulting, small magnetic fields.
Venus	Shrouded in clouds, undulating surface with highlands, plains, lowlands and craters.
Earth	Ocens of water filling lowlands between continents, unique in supporting life, magnetic field.
Mars	Cratered uplands, lowland plains, volcanic regions.
Jupiter	Covered by clouds, dark ring of dust, magnetic field.
Saturn	Several cloud layers, magnetic field, thousands of rings.
Uranus	Layers of cloud and mist, magentic field, some rings.
Neptune	Unable to detect from earth.
Pluto	Unable to detect from earth

It is produced from the input

```
\begin{center}
  \begin{tabular}{lp{.8\linewidth}}
    Planet  & Features\\[5pt]
    Mercury & Lunar like crust, crustal faulting, small magnetic
              fields.\\

    Venus   & Shrouded in clouds, undulating surface with highlands,
               plains, lowlands and craters.\\
    Earth   & Ocens of water filling lowlands between continents,
              unique in supporting life, magnetic field.\\
    Mars    & Cratered uplands, lowland plains, volcanic regions.\\
    Jupiter & Covered by clouds, dark ring of dust, magnetic field.\\
    Saturn  & Several cloud layers, magnetic field, thousands
              of rings.\\
    Uranus  & Layers of cloud and mist, magentic field, some rings.\\
    Neptune & Unable to detect from earth.\\
    Pluto   & Unable to detect from earth
  \end{tabular}
\end{center}
```

Here the specification `p{6cm}` shows that in a "paragraph box" of width 6 cm. In a p-type column, if a `\raggedright` or `\centering` is given, then we can induce explicit line breaks *within that column* by the `\\` command. If such commands are used in the last column of a row, then the command `\tabularnewline` should be used to terminate that row as in this example:

```
\begin{center}
  \begin{tabular}{lp{6cm}}
    Planet  & Features\tabularnewline[8pt]
    Mercury & \raggedright Lunar like crust\\
                           Crustal faulting\\
                           Small magnetic fiels\tabularnewline[3pt]
    Venus   & \raggedright Shrouded in clouds\\
                           Undulating surface\tabularnewline[3pt]
    Earth   & \raggedright Ocens of water\\
                           Unique in supporting life\\
                           Magnetic field\tabularnewline[3pt]
    Mars    & \raggedright Cratered uplands\\
```

```
                        Lowland plains\\
                        Volcanic regions\tabularnewline[3pt]
    Jupiter & \raggedright Covered by clouds\\
                        Dark ring of dust\\
                        Magnetic field\tabularnewline[3pt]
    Saturn  & \raggedright Several cloud layers Magnetic field\\
                        Thousands of rings\tabularnewline[3pt]
    Uranus  & \raggedright Layers of cloud and mist\\
                        Magentic field\\
                        Some rings\tabularnewline[3pt]
    Neptune &           Unable to detect
                               from earth\tabularnewline[3pt]
    Pluto   &           Unable to detect
                               from earth\tabularnewline[3pt]
  \end{tabular}
\end{center}
```

This produces the output below

Planet	Features
Mercury	Lunar like crust Crustal faulting Small magnetic fiels
Venus	Shrouded in clouds Undulating surface
Earth	Ocens of water Unique in supporting life Magnetic field
Mars	Cratered uplands Lowland plains Volcanic regions
Jupiter	Covered by clouds Dark ring of dust Magnetic field
Saturn	Several cloud layers Magnetic field Thousands of rings
Uranus	Layers of cloud and mist Magentic field Some rings
Neptune	Unable to detect from earth
Pluto	Unable to detect from earth

Note that the last two lines don't need a `\raggedright` command, since there are no explicit linebreaks in them.

A table usually contains horizonntal and vertical lines separating the rows and columns. These can also be produced in the `tabular` environment. For example, the first table we saw above can be typeset as

Planet	Diameter(km)
Mercury	4878
Venus	12104
Earth	12756
Mars	6794
Jupiter	142984
Saturn	120536
Uranus	51118
Neptune	49532
Pluto	2274

by the input

```
\begin{center}
  \begin{tabular}{|l|r|}
    \hline
    Planet  & Diameter(km)\\
    \hline
    Mercury & 4878\\
    ..............
    Pluto   & 2274\\
    \hline
  \end{tabular}
\end{center}
```

Do you see what produced the vertical and horizontal lines? Instead of the specification `{lr}` used earlier, we now have `{|l|r|}` The character `|` causes a vertical line to be drawn at the specified location, running down the entire height of the table. (Two `|`'s in succession produce a double vertical lines.) An `\hline` command after a row draws a horizontal line after that row, running along the entire width of the table. (Again, two `\hline`'s in succession producea double horizontal line.) Note also that because of the last `\hline` , we should give a line termination command `\\` at the end of the last row also.

Now suppose we want to produce something like this

Planet	Distance from sun (km)	
	Maximum	Minimum
Mercury	69400000	46800000
Venus	109000000	107600000
Earth	152600000	147400000
Mars	249200000	207300000
Jupiter	817400000	741600000
Saturn	1512000000	1346000000
Uranus	3011000000	2740000000

Here, there are three columns and the entry `Distance from the sun (km)` is to span the the last two columns below it. The command `\multicolumn` does the trick as shown below

```
\begin{center}
  \begin{tabular}{lrr}
    Planet  & \multicolumn{2}{c|}{Distance from sun (km)}\\
            & Maximum    & Minimum\\
```

```
    Mercury & 69400000   & 46800000\\
    Venus   & 109000000  & 107600000\\
    Earth   & 152600000  & 147400000\\
    Mars    & 249200000  & 207300000\\
    Jupiter & 817400000  & 741600000\\
    Saturn  & 1512000000 & 1346000000\\
    Uranus  & 3011000000 & 2740000000\\
  \end{tabular}
\end{center}
```

The entry `\multicolumn{2}{c}{Distance from sun (km)}` indicates that the *item* within the last set of braces is to span *two* columns as specified by the `2` within the first set of braces. The entry `c` within the second set of bracesindicates that this text is to be *centered* within the column. Thus the general form of the command is

`\multicolumn{`*num*`}{`*pos*`}{`*item*`}`

where *num* is the number of columns to be spanned, *pos* is the position of the item within the column and *item* is the text of the item. Note also that the input for the second row *starts* with an `&` character. This is because there is no entry in the first column of the second row.

Now what if you want

Planet	Distance from sun (km)	
	Maximum	Minimum
Mercury	69400000	46800000
Venus	109000000	107600000
Earth	152600000	147400000
Mars	249200000	207300000
Jupiter	817400000	741600000
Saturn	1512000000	1346000000
Uranus	3011000000	2740000000
Neptune	4543000000	4466000000
Pluto	7346000000	4461000000

Here the first few lines and the last lines of the input are as below (the other lines are the same as in the previous example).

```
\begin{center}
  \begin{tabular}{|l|r|r|}
    \hline
    Planet & \multicolumn{2}{c|}{Distance from sun (km)}\\
    \cline{2-3}
           & Maximum   & Minimum\\
    \hline

    ..............................

    \hline
  \end{tabular}
\end{center}
```

Note that the position specifier in the `\multicolumn` command here is `c|`. This has to do with the way the environment splits the column specification into various columns.

For example, the specification `|l|r|r|` in this exaple is split into `|l|`, `r|` and `r|` and the `\multicolumn{2}` command resets the last two columns. In particular, the final `|` gets reset and we'll have to explicitly supply it in the position specification of the `\multicolumn` command as `c|`.

Note also the command `\cline{2-3}` after the first row. This draws a horizontal line from the second to the third column. In general the command `\cline{`*i*`-`*j*`}` draws a horizontal line from the ith column to the jth column.

Another feature of the `\multicolumn` command is that with `\multicolumn{1}` we can override the position specification of any column set at the beginning of the environment. For example, consider the input below

```
\begin{center}
  \begin{tabular}{|l|r|r|}
    \hline
           & \multicolumn{2}{p{3.5cm}|}%
               {\centering Distance from sun \\ (million km)}\\
    \cline{2-3}
    \multicolumn{1}{|c|}{Planet}
           & \multicolumn{1}{c|}{Maximum}
                  & \multicolumn{1}{c|}{Minimum}\\
    \hline
    Mercury & 69.4   & 46.8\\
    Venus   & 109.0  & 107.6\\
    Earth   & 152.6  & 147.4\\
    Mars    & 249.2  & 207.3\\
    Jupiter & 817.4  & 741.6\\
    Saturn  & 1512.0 & 1346.0\\
    Uranus  & 3011.0 & 2740.0\\
    Neptune & 4543.0 & 4466.0\\
    Pluto   & 7346.0 & 4461.0\\
    \hline
  \end{tabular}
\end{center}
```

It produces the output below

Planet	Distance from sun (million km)	
	Maximum	Minimum
Mercury	69.4	46.8
Venus	109.0	107.6
Earth	152.6	147.4
Mars	249.2	207.3
Jupiter	817.4	741.6
Saturn	1512.0	1346.0
Uranus	3011.0	2740.0
Neptune	4543.0	4466.0
Pluto	7346.0	4461.0

Note that even though `\centering` is used in the last column of the first row, no `\tabularnewline` is required to terminate this row, since the scope of the `\centering` is limited by the `\multicolumn`.

By the way, do you feel that the tables we've been produced look a bit cramped? A bit crowded vertically? Well, you can create a bit more room between *rows* by redefining the value of `\arraystretch`. By default, it's value is 1 and if you set it to a number k, then the interrow space is increased k-fold. Thus the input of the last example with the command

```
\renewcommand{\arraystretch}{1.2}
```

after the `\begin{center}` produces

Planet	Distance from sun (million km)	
	Maximum	Minimum
Mercury	69.4	46.8
Venus	109.0	107.6
Earth	152.6	147.4
Mars	249.2	207.3
Jupiter	817.4	741.6
Saturn	1512.0	1346.0
Uranus	3011.0	2740.0
Neptune	4543.0	4466.0
Pluto	7346.0	4461.0

Next let's see how we produce a table like the one below

Height (cm)	Ideal weight (kg)
155	53.5–64
160	56–67
165	59–71
170	62.5–75.5
175	66–79
180	70–83.5
185	71.5–86.5
190	78–92.5

Here we want all the dashes in the second column to be vertically aligned, so that we must set them in a separate column; but then there should be no space between the numbers and the dashes connecting them. In such cases we can use the `@` command in the column specification as below

```
\begin{center}
  \begin{tabular}{|c|r@{--}l|}
    \hline
    Height & \multicolumn{2}{c|}{Ideal weight}\\
    (cm)   & \multicolumn{2}{c|}{(kg)}\\
    \hline
    155 & 53.5 & 64\\
    160 & 56   & 67\\
    ..............
    190 & 78   & 92.5\\
```

```
    \hline
  \end{tabular}
\end{center}
```

Here the specification `r@{--}l` indicates that there should be a right aligned column and a left aligned column with a – in between each pair of entries in these columns *without the intercolumn space* the `tabular` environment leaves by default between every pair of columns. Note that this incidently saves us the trouble of repeatedly typing `--`. You can also add some space producing commands within the braces after the `@` command to produce that much space between the columns on either side of it.

VII.2.1. Enhancements to the `tabular`

There are many packages which provide further facilities in forming tables. We'll discuss a couple of such packages here.

VII.2.2. The **array** package

Look at the tables below

Planet	Mean distance from sun (km)
Mercury	58100000
Venus	108300000
Earth	150000000
Mars	228250000
Jupiter	779500000
Saturn	1429000000
Uranus	2439000000
Neptune	4504500000
Pluto	5903500000

Planet	Mean distance from sun (km)
Mercury	58100000
Venus	108300000
Earth	150000000
Mars	228250000
Jupiter	779500000
Saturn	1429000000
Uranus	2439000000
Neptune	4504500000
Pluto	5903500000

The one on the right looks nicer, doesn't it? It was produced using the column specifier `m` available in the array package. To produce this table, we must first load the array package by the ususl `\usepackage{array}` in the preamble and then type

```
\begin{tabular}{|l|r|}
 \hline
 \multicolumn{1}{|m{1.5cm}|}{\centering Planet}
           &\multicolumn{1}{m{2.3cm}|}%
             {\centering Mean distance from sun \\ (km)}\\
  \hline
   Mercury & 58100000\\
   ...................
   Pluto   & 5903500000\\
   \hline
 \end{tabular}
```

The `m{`*wd*`}` specifier produces a column of width *wd* just like the `p` specifier, but with the text aligned vertically in the middle unlike the `p` specifier which aligns the text with the topline. (The table on the left, incidently, was produced by the same input as above but with `p` instead of `m`).

Another interesting feature of the array package is the >{*decl*} command which can be used before a column specifier. It inserts *decl* directly in front of the column. For example look at the input below

```
\begin{center}
  \begin{tabular}{|>{\bfseries}l|r|}
    \hline
    \multicolumn{1}{|m{1.5cm}|}{\centering Planet}
          &\multicolumn{1}{m{2.3cm}|}%
            {\centering Mean distance from sun \\ (km)}\\
    \hline
    Mercury & 58100000\\
    Venus   & 108300000\\
    Earth   & 150000000\\
    Mars    & 228250000\\
    Jupiter & 779500000\\
    Saturn  & 1429000000\\
    Uranus  & 2439000000\\
    Neptune & 4504500000\\
    Pluto   & 5903500000\\
    \hline
  \end{tabular}
\end{center}
```

which produces the output

Planet	Mean distance from sun (km)
Mercury	58100000
Venus	108300000
Earth	150000000
Mars	228250000
Jupiter	779500000
Saturn	1429000000
Uranus	2439000000
Neptune	4504500000
Pluto	5903500000

The array package also has a ! command which works just like the @ command, but whch does not suppress the intercolumn space.

VII.2.3. The multirow package

Look again at the table in 68. Wouldn't it be nice if the entry "Planet" in the first column is vertically aligned with the center of the two rows in the next column as below?

Planet	Distance from sun (million km)	
	Maximum	Minimum
Mercury	69.4	46.8
Venus	109.0	107.6
Earth	152.6	147.4
Mars	249.2	207.3
Jupiter	817.4	741.6
Saturn	1512.0	1346.0
Uranus	3011.0	2740.0
Neptune	4543.0	4466.0
Pluto	7346.0	4461.0

The package multirow is what we need to do this painlessly. It has a command

\multirow{*num*}{*wd*}{*item*}

where *num* is the number of rows to be spanned, *wd* is the width of this column and *item* is the text of the item in this column. This can be used as in the following example

```
  \begin{center}
    \begin{tabular}{|l|r|r|}
    \hline
    \multirow{3}{1.5cm}{Planet}
            & \multicolumn{2}{p{3.5cm}|}%
                 {\centering Distance from sun \\ (million km)}\\
    \cline{2-3}
            & \multicolumn{1}{c|}{Maximum}
                     & \multicolumn{1}{c|}{Minimum}\\
    \hline
    Mercury & 69.4   & 46.8\\
    Venus   & 109.0  & 107.6\\
    Earth   & 152.6  & 147.4\\
    Mars    & 249.2  & 207.3\\
    Jupiter & 817.4  & 741.6\\
    Saturn  & 1512.0 & 1346.0\\
    Uranus  & 3011.0 & 2740.0\\
    Neptune & 4543.0 & 4466.0\\
    Pluto   & 7346.0 & 4461.0\\
    \hline
  \end{tabular}
\end{center}
```

But this code does not produce the table above, but only

Planet	Distance from sun (million km)	
	Maximum	Minimum
Mercury	69.4	46.8
Venus	109.0	107.6
Earth	152.6	147.4
Mars	249.2	207.3
Jupiter	817.4	741.6
Saturn	1512.0	1346.0
Uranus	3011.0	2740.0
Neptune	4543.0	4466.0
Pluto	7346.0	4461.0

The trouble is that though the entry "Planet" is vertically centered in its column, it is not horizontally centered. The horizontal alignment is controlled by the command `\multirowsetup` and this is by default st to `\raggedright`. So all that is needed to get the beautiful table seen at the beginning of this section is to add the line

```
\renewcommand{\multirowsetup}{\centering}
```

at the beginning of the code above.

VII.2.4. `tabbing` vs. `tabular`

Let's take a quick look at the pros and cons of the `tabbing` and `tabular` environments.

- The `tabbing` environment can be typeset only as a separate paragraph, while the `tabular` environment can be placed anywhere in text, even inside Mathematics.
- The `tabbing` environment can span multiple pages, but the `tabular` environment cannot.
- `tabbing` environments cannot be nested, while `tabular` environments can be nested to any number of levels.

VII.2.5. Multipage tables—The package longtable

As we have noted, we cannot create table spanning more than one page using the `tabular` environment. But the package longtable by David Carlisle can do this and it has quite a few other tricks also. To use this package, load it as usual with the command `\usepackage{longtable}` in the preamble and then to produce a no-frills "longtable" just use the commands `\begin{longtable} ... \end{longtable}` instead of the `\begin{tabular} ... \end{tabular}` commands. We can use footnotes and the `\newpage` commands inside the `longtable` environment. If the package array is also loaded, its extra features can be used.

Apart from this, this package has provisions to specify *at the start of the input* the following items

- the rows that should appear at the *top of the table*; the input for these to be terminated by `\endfirsthead`
- the rows that should appear in *every page after the first*, such input terminated by `\endhead`
- those at the *bottom of every page*, the input terminated by `\endfoot`

- those rows at the *very end* of the table, terminated by `\endlastfoot`

These are illustrated in the (long!) table below.

Science and Technology in the Twentieth Century

Year	Event
1900	Max Planck proposes quantum theory
	Publication of Sigmund Freud's *The Interpretation of Dreams*
1901	Discovery of principal blood groups
	Guglielmo Marconi transmits wireless signals across the atlantic
1903	Wright brothers make their first flight
1905	Albert Einstein presents Special Theory of Relativity
1911	Ernest Rutherford proposes theory of atomic structure
1912	Victor Hess discovers cosmic rays
1916	Albert Einstein presents general Theory of Relativity
1920	Radio broadcasting begins
1926	John Logie Baird demonstrates television
1928	Alexander Fleming discovers penicillin
1933	Discovery of polythene
1934	Discovery of nuclear fission
1938	Discovery of nylon
1940	Plutonium obtained by bombardment of uranium
1942	Construction of first nuclear reactor
1946	Construction of first electronic digital computer
1947	First supersonic flight
	Invention of the transistor
1951	Nuclear power stations introduced
1953	James Watson and Francis Crick show DNS molecule structure
1956	Contraceptive pill introduced
1957	Launch of the first space satellite (*Sputnik 1*)
1959	First photograph of the dark side of the moon (*Luna 3*)
...	.....
...	.....
...	.....
...	.....
...	.....
...	.....
...	.....
...	.....
...	.....
...	.....
...	.....
...	.....
1961	Yuri Gagarin becomes first man in space (*Vostok 1*)
1966	First lunar soft landing (*Luna 9*)
1967	Discovery of pulsars
1968	First manned lunar orbit (*Apollo 8*)
1969	First man on moon (Neil Armstrong)
1972	Pocket calculator introduced

continued on the next page

Science and Technology in the Twentieth Century (*continued*)

Year	Event
1974	First 'test-tube babies'
1977	Launch of *Voyager* missions to outer spce
1983	IBM personal computer launched
1986	Hailey's comet intercepted
1997	Cloning of "Dolly" the sheep
2000	Decoding of 90% of human genome completed

Source : *The Cambrige Factfinder*

Part of the code to produce this is given below.

```
\renewcommand{\arraystretch}{1.2}
\begin{longtable}{|c|l|}
  \multicolumn{2}{c}%
    {\textbf{Science and Technology in the Twentieth Century}}\\[5pt]
  \hline
  \multicolumn{1}{|c|}{\sffamily Year}
  &\multicolumn{1}{|c|}{\sffamily Event}\\
  \hline
  \endfirsthead
   \multicolumn{2}{c}%
      {\textbf{Science and Technology in the Twentieth Century}
      (\textit{continued})}\\[5pt]
  \hline
  \multicolumn{1}{|c|}{\sffamily Year}
  &\multicolumn{1}{|c|}{\sffamily Event}\\
  \hline
  \endhead
  \hline
  \multicolumn{2}{r}{\small\itshape continued on the next page}\\
  \endfoot
  \hline
  \multicolumn{2}{r}{\small Source\,:\,\itshape The Cambrige Factfinder}
  \endlastfoot
  1900 & Max Planck proposes quantum theory\\
  ..........................................
  2000 & Decoding of 90\% of human genome completed\\
  \hline
\end{longtable}
```

VII.2.6. And that's not all!

There are many more packages which help to produce tables of various requirements. Be sure to check out the pakages tabularx, delarray, dcolumn and hhline.